

Proactivity in Scaffolding Comprehension and Production in Real-Time Turn-Taking: A Case Study of Bridging Communication Gaps for Non-Native Speakers

Wanqing Psyche He
Department of Information Science
Cornell University
Ithaca, New York, USA
wh385@cornell.edu

Susan R. Fussell
Department of Information Science
Cornell University
Ithaca, New York, USA
sfussell@cornell.edu

Abstract

Predictive processing is cognitively challenging in rapid turn-taking, which can induce miscomprehension and misalignment, especially for those with more limited domain-specific knowledge, language proficiency, and cultural experience. We designed a proactive system, XPLAIN, that offloads human prediction to assistive tools, thereby mitigating communication gaps in turn-taking. In a lab-based Wizard-of-Oz study, we evaluated its features (i.e., lexical clarifications, idea and content suggestions, and topic summaries) in scaffolding dyadic virtual meetings on attested theoretical grounds. As expected, XPLAIN improved efficiency and participation in virtual meetings, and promoted inclusivity and active participation, subject to individual differences in language proficiency, personality, and prior experience with AI. This study provides insights for design architects on how to boost smoothness and efficiency in real-time cognitive processing with psychological support for confidence, assurance, engagement, and autonomy. It also advocates for customizing proactive systems aside from the 'one-size-fits-all' approach to internalize user characteristics and background.

CCS Concepts

• **Human-centered computing** → **Laboratory experiments; Usability testing.**

Keywords

Proactive systems; AI-mediated communication; Conversation scaffolding; Knowledge gap; Bilingual speakers

ACM Reference Format:

Wanqing Psyche He and Susan R. Fussell. 2025. Proactivity in Scaffolding Comprehension and Production in Real-Time Turn-Taking: A Case Study of Bridging Communication Gaps for Non-Native Speakers. In *Companion of the Computer-Supported Cooperative Work and Social Computing (CSCW Companion '25)*, October 18–22, 2025, Bergen, Norway. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3715070.3749273>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CSCW Companion '25, Bergen, Norway
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1480-1/25/10
<https://doi.org/10.1145/3715070.3749273>

1 Introduction

Turn-taking represents one of communication's most compressing processes, requiring speakers to simultaneously comprehend incoming speech, predict appropriate content and timing, and formulate responses [39]. Disfluencies (i.e., repetitions, hesitations, and repairs in speech [8, 24]) are prevalent in face-to-face interactions (F2F) [7] but particularly problematic in computer-mediated communication (CMC) that deteriorate comprehension, efficiency, and engagement [23, 42, 58]. A critical yet overlooked source of disfluencies in CMC is content prediction—the ability to anticipate upcoming linguistic elements [2, 50, 51]. Impoverished visual and auditory cues necessary for prediction impose substantial challenges for smooth and efficient turn-taking, particularly detrimental for non-native speakers (NNS) whose cue integration is often delayed and less efficient due to more limited language proficiency and cultural awareness [15, 22, 29, 30, 57].

Existing technologies supporting communication for NNS often overlook the time-sensitive need of comprehension, prediction, and production in turn-taking and treat them as uniform challenges requiring generic solutions [23, 40, 43, 48, 52]. Therefore, the goal of this study is to investigate the effectiveness of offloading human prediction to artificial intelligence (AI) tools to aid information processing and minimize communication gaps for NNS in CMC. We introduce XPLAIN, a proactive system designed to scaffold real-time turn-taking by anticipating, avoiding, and addressing conversational gaps that impede effective communication and participation via tailored support:

- (1) **Comprehension:** pre-identifying linguistic, cultural, and domain-specific knowledge gaps, including complex terms, slang, idioms, acronyms, concepts, and cultural references to minimize potential misunderstanding [13].
- (2) **Production:** pre-activating contextually relevant information and associative relationships of relevant topics to foster early preparation of conceptualization, idea formation, and output construction to reduce the planning effort.
- (3) **Alignment:** promoting real-time conversation alignment to maintain common ground by reducing the pressure of short-term memory and improving the validity of information exchange [55].

We designed three contextual features—lexical clarifications, idea and content suggestions, and topic summaries—each directly targeted at mitigating the specified gaps above. Through a Wizard-of-Oz (WoZ) study followed by in-depth interviews, our results showed that integrating such features into online meetings can

enhance efficiency, foster inclusivity and engagement, and boost psychological safety and confidence. XPLAIN effectively acted as a cognitive scaffold by managing excessive load to facilitate processing and retention. We also identified variations in language proficiency, personality, and prior experience with AI, calling for customized systems that better balance proactive assistance with user autonomy.

This work contributes to the current research by 1) applying psycholinguistic theories of prediction to practical CMC system design; 2) supporting both comprehension and production phases in real time turn-taking, demonstrating the value of differentiating communication gaps with distinct theoretical grounds; and 3) offering empirically grounded design principles for developing inclusive tools that enhance participation without compromising flow smoothness, user agency, or active engagement.

2 Related Works

2.1 Human prediction vs. model prediction

Prediction serves as the cornerstone of human language processing, enabling efficient real-time communication by anticipating linguistic elements before they occur with reduced cognitive load to facilitate rapid responses [15, 32, 36, 46, 54]. Predictive coding plays a crucial role in resolving communication disfluency, enabling the brain to anticipate and quickly adjust to incoming linguistic input, thus enhancing fluency and coherence. These predictions rely on local conversational cues [10, 14, 27, 45, 47, 53], and global contextual cues such as topic, shared knowledge, and social relationship [46, 59]. Cue-impooverished CMC can amplify prediction errors at various linguistic levels, leading to disfluencies that disrupt communication [29, 30, 39, 56]. Although existing systems [23, 40, 43, 48, 52] provide general support for NNS in CMC, they typically treat language barriers as uniform challenges. Linguistic competence is multifaceted, encompassing comprehension, planning, and production, each requiring different forms of assistance to effectively scaffold turn-taking.

2.2 Proactive assistance in real-time

The recent shift from reactive to proactive systems has enabled a critical advancement in supporting fluid conversation [4, 5]. Strategically offloading tasks to assistive AI tools can close different types of conversation gaps [3, 18–20, 34, 41, 49, 61]. Reactive prompting can be challenging for users facing language barriers or unfamiliar terminology [28]. Alternatively, outputs that align with the natural rhythm of conversation have been found to improve user acceptance and mitigate disruptions [35, 44]. Proactivity not only enhances user engagement but also fosters autonomy, allowing users to make informed decisions without over-reliance on external information [1, 9, 34]. With contextually embedded assistance, proactive systems can manage cognitive demands and provide aid without interrupting the natural flow of turn-taking.

Such advancements are highly valued in collaborative settings, where real-time efficiency and seamless communication are critical to participation, especially for NNS with limited proficiency [23, 25, 60]. HCI systems can mitigate the processing challenges faced by NNS that impede their effective communication [38], enabling equitable participation from education to professional collaboration,

fostering more productive, engaging, and inclusive interactions for NNS with communication barriers.

Built upon these objectives, this study evaluates both prediction theory to mitigate communication challenges in CMC and the functionality of proactive prediction in AI assistance. We expected advanced user experience in confidence and engagement, subject to prior experience and language proficiency while being effectively supported by the predictive tools.

3 Method

3.1 Participant

Twenty-seven NNS of English or Japanese (M=8; F=19; average age=21.12) were recruited via the university SONA for extra course credits. Some (N=7) grew up and received education in an English-speaking country before 12 years old. Most (N=17) lived in an English-speaking country between 12 and 18 years old. A few (N=3) just arrived for college education. Participants were denoted by their native language (C=Chinese, G=German, H=Hindi, K=Korean, P=Persian, U=Urdu) and years spent in an English-speaking country (e.g., C3-7 = 3rd Chinese-speaking participant, 7 years).

3.2 Tool design description

We ran a WoZ study with Figma with three essential features: clarification, suggestion, and summary, as in Figure 1. To reduce the adaptation effort, two interactive components were included: 1) a caption, where each green word is clickable with a pop-up box above that clarifies the term, and 2) a sidebar, where each feature pops up separately in a temporal order. The tool was designed to be perceived as a fully automated AI agent with all outputs proactively generated when needed during the conversation without user prompting.

3.3 Materials

The collaborative task conversation was to design menu items for a Japanese American fusion restaurant in College Town. The procedure was standardized by making the confederate part fully pre-scripted to consistently guide the conversation flow. Validated via pre-testing (N=3), this theme ensured low topic unfamiliarity for target participants from a third culture and enabled naturalistic exchanges via open-ended questions. Clarifications included 5 slang, 5 acronyms, 6 idioms, 6 complex words, and 6 cultural food items with images. The words within each category varied in relative complexity but roughly matched in lexical frequency in COCA [16]. All were positioned in the first half of the caption to ensure enough processing time. The maximal length of clarifications was restricted to twenty words. Six idea suggestions (3 American and 3 Japanese items with images) and four sentence suggestions (with bidirectional opinions to avoid bias) auto popped up immediately after questions. The length of suggestions was restricted to one full sentence for each idea or structure. Four summaries, formatted in short bullet points with high information density, auto popped up immediately after topic transitions. All features were only applied to the zoom account of the participants and distributed evenly in the conversation to balance the cognitive load. More detailed materials can be found on Open Science Framework (osf.io/yc6dh).

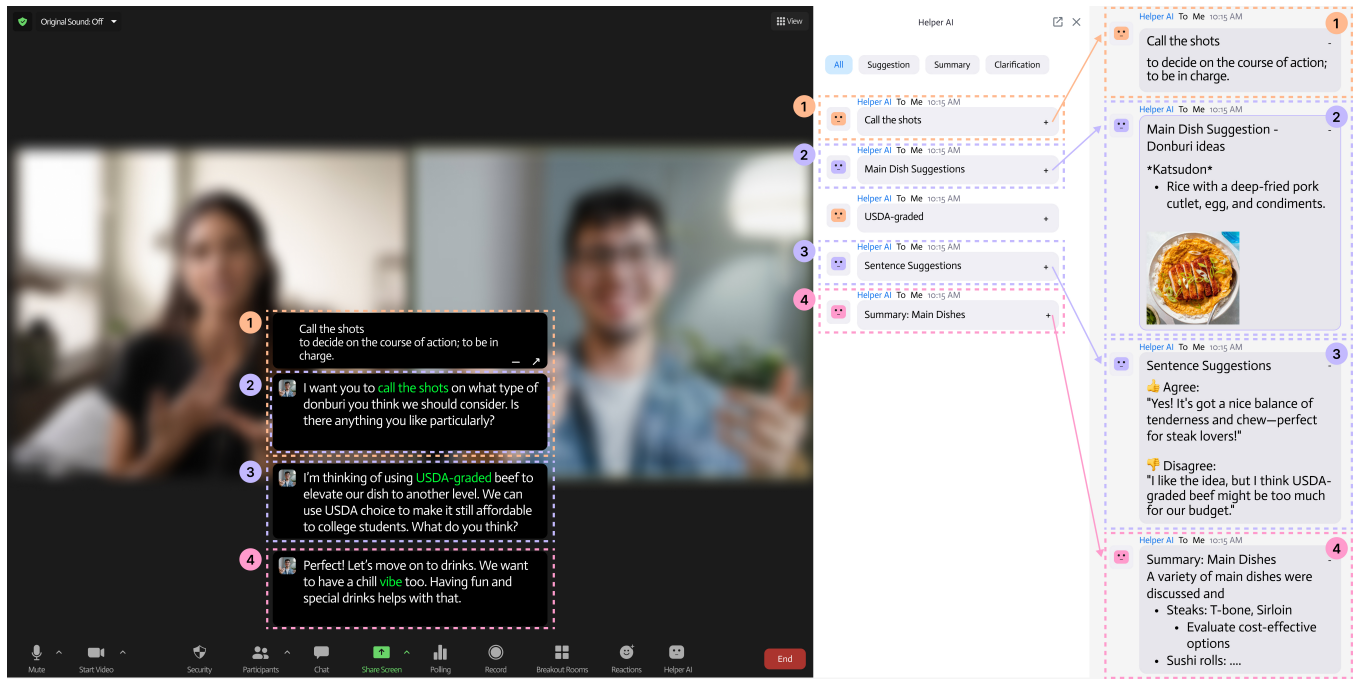


Figure 1: Zoom interface with XPLAIN; 1: clarification box popped up above caption once clicked with minimal eye movement track, saved as minimized in the side bar for future reference; 2: idea suggestion; 3: full-sentence suggestion; 4: summary. 2 3 4 all popped up as expanded windows. Only one feature expands at a time to avoid visual overload. (N.B. the caption stacks were for illustration purposes: only one caption box stayed at the bottom position as the conversation progressed.)

A semi-structured interview protocol was developed, asking participants to reflect on their communication experience: overall comprehensibility of the conversation, perceived effectiveness of each feature, change in behavioral engagement, change in communication effort, comfort with the interface, alongside their background and prior experience with AI.

3.4 Procedure

Two laptops joined over Zoom and connected in TeamViewer. Zoom windows were pinned on top of the Figma page's placeholder to mimic a real meeting integrated with XPLAIN. The confederate monitored and progressed the conversation via TeamViewer. After consent, participants joined a demo session to interact with features (3 mins). In the official session (15 mins), the confederate had a natural conversation with users. Then, they shared feedback via an interview (15 mins).

4 Results

All interviews were recorded and transcribed by Otter.ai and manually reviewed for errors. Using an inductive and interpretive approach [26], general themes were first identified from the transcripts, with high-level themes and their relationships further constructed. Codes were refined through an iterative process of creating and combining themes.

4.1 Overall enhanced experience during the conversation

Participants (14/27) consistently highlighted clarification in maintaining conversational flow and enhancing comprehension in Zoom meetings. It was found effective in reducing the cognitive burden of navigating unfamiliar and domain-specific language: *"In lectures where complex terms come up often, having clarifications pop up would definitely help me follow along better."* (K3-2). Late NNS found it helpful with culturally specific terms (age of acquisition = 6–12).

Participants (9/27) reported feeling more confident engaging in discussions. Among all features examined, clarification tools emerged as the most appreciated for supporting immediate assistance: *"I'm able to understand more new concepts on spot, so I can engage the conversation more... I can probably [ask] deeper question regarding whatever we're talking about that time"* (C6-3). Temporally aligned suggestions helped with speech planning processes that *"inspire me by providing relevant ideas while allowing me to contribute my own thoughts."* (C12-6). Images were perceived as exceptionally efficient, *"especially for cases like food items where my brain does not process the word but understands the image immediately."* (C9-5). Participants who reported increased engagement from XPLAIN often struggled with cultural or linguistic barriers, as C2-3 mentioned, *"sometimes I don't really know what I can give, because I don't really have that knowledge in my mind."* XPLAIN reduced their anxiety of awkwardness and gaps in formulating questions.

4.2 Perceived effectiveness of individual features: whether and how they were found effective

Participants (11/27) highlighted how XPLAIN acted as an effective time-saver that reduced the need for manual searches with retained attention, ranging from 30% to 80% reported boost in comprehension efficiency. Visual features designed with minimized effort of gaze shift were valued. There was also a behavioral change in verification with enhanced psychological safety, *"I would check the text in green just in case I have the wrong definition of it in mind and check if I was understanding the question correctly."* (K2-4). However, six participants reported experiencing temporal overload given the compressed nature of turn-taking: *"you take time to look at the explanation at the same time, maybe the other person is still speaking, but you will miss what they just say."* (C4-3).

Opinions on suggestions were more divided. They effectively reduced pauses and effort of idea generation. Meanwhile, concerns about over-reliance on AI with reduced thinking processes were raised. More proficient speakers (5/27) felt they had limited originality or *"may manipulate thinking"* (C8-2). Some felt AI-generated output may not be trustworthy: *"I may not give the most accurate definition, but given I have a resource, how can [it] be a more accurate definition?"* (C6-3).

Summaries were perceived as particularly useful in structured settings such as lectures or formal meetings. Participants (8/27) valued it for capturing key points as a real-time note-taker: *"it consolidates information... aids in reviewing academic lectures, especially for difficult professors."* (C5-5). They were found to be excessive in unstructured conversations as *"they don't address immediate conversational needs. They might be useful for project documentation instead."* (C12-6).

Participants also expressed the desire for customizable features based on situational contexts or personal preferences: *"if there's like a button setting... on the radical side or conservative side' in terms of helping the definition."* (C6-3).

4.3 Individual differences in personality, prior experience, and trust in AI tools

Participants (9/27) with a naturally curious disposition were more deeply engaged with features, exploring beyond their immediate needs. U1-1 explained, *"Clarifications I didn't need... But I was so curious to see what sort of how it may help me, so I did click around."* This interaction highlights the potential for AI tools to inspire creativity and broaden understanding when approached with an open and inquisitive mindset.

A few participants (5/27) who identified as independent thinkers expressed skepticism toward suggestions. They were more proficient speakers and experienced users of AI tools, viewing reliance on the tool as potentially diminishing their personal input: *"I still want to vet my story and use my own opinion in writing over what they gave me."* (C7-9). In contrast, some participants (4/27) rather compromised their originality for the high-quality output: *"this response is better than anything I would have given, even though it didn't feel as authentic."* (H1-9).

Participants exhibited varying levels of confidence and communication styles relative to their degree of language anxiety. Some

(8/27) expressed hesitation to ask others for clarification during meetings, citing fears of interruption and embarrassment. Clarifications were deemed effective in alleviating this anxiety that was more pronounced among younger participants, especially in more demanding settings: *"I will use it in professional meetings... you don't want to spend too much time asking clarifications... to slow everyone's sound during those meetings."* (C12-6). Some (5/27) hesitated to use publicly in formal settings due to concerns of professional readiness. K3-2 emphasized that *"for interviews, it would be really helpful to bring up words that I have no idea about, and if I asked them what it was, it would negatively affect me."*

4.4 The intuitive UI design of XPLAIN

Participants (8/27) found these features required minimal effort to access and seamlessly integrated into their workflow: *"Clicking on green text for definitions was helpful because it was right above it, without switching tabs."* (C7-9). Six participants new to AI tools showed rapid adaptation: *"It was easy to pick up even though I've never used anything like this before—it felt natural."* (U1-1). However, six users found the UI cluttered when they expanded multiple features simultaneously: *"It offers more information than I often need; sometimes feels overwhelming or redundant."* (P1-4).

5 Discussions and Future Directions

Overall, XPLAIN's contextualized features integrated into Zoom were perceived positively to enhance comprehension, efficiency, and engagement. The timely yet private nature emerged as a pivotal advantage, allowing NNS to resolve misunderstandings without publicly revealing their confusion to foster a supportive environment for all users. However, there are downsides such as cognitive overload from the proactive pop-ups, distrust or overdependence on AI-generated outputs. In line with our hypotheses, these findings highlight areas for improvement in future designs: balancing functionality with user autonomy and preference is critical without introducing new barriers.

First, system adaptability should address individual differences for effective human-computer collaboration [21]. Participants valued individualized features to reduce redundancy in real-time processing. However, it is challenging to gauge the knowledge gaps of users to generate tailored clarifications, especially for jargon and acronyms. Using 1) models trained on prior conversations, 2) gathering language, cultural, and professional experience, and 3) constructing a personal knowledge base for industry/subject-specific terms by users will advance tool effectiveness through extensive knowledge management. Customizable options in the tool settings can also allow users to adjust the frequency and depth of information for different meeting scenarios without feeling overwhelmed by complexity or underwhelmed by simplicity.

Another central need is the flexibility of supporting different communication styles and personality traits in line with sociocultural theories of situated learning, emphasizing that cultural and educational contexts significantly shape the adoption and use of technology [37]. Less confident users could benefit from interaction in a private, unobtrusive manner, alleviating the social anxiety and the stigma associated with seeking help [22, 23, 60] to elevate

confidence, participation, and inclusivity. More confident and proactively engaged users might benefit more from deeper analytical support with exploratory features for public brainstorming and idea generation that encourage a more dynamic and creative experience.

While suggestions improved idea fluency, concerns about reduced cognitive engagement from offloading [12, 17] necessitate designs that promote active information processing to avoid overreliance [6, 31, 33]. For low-proficiency NNS, adaptive suggestion timing, delaying output by 3–5 seconds post query, and reduced output density (e.g., sentence template) could encourage initial idea formulation. High-proficiency NNS could benefit from explanatory or comparative scaffolding (e.g., multiple options with confidence scores) for critical evaluation over passive adoption to maintain autonomy [1, 9, 11].

Real-time summaries were perceived as more effective in structured meetings with alignment needs. To address this, topic-density triggers should activate summarization only if too many subtopics emerge or too many clarifications pop up within a short time span, signaling potential cognitive overload. In structured meetings, summaries should also have higher frequency and depth of information, publicly shared across interlocutors for real-time alignment and liability purposes.

The time sensitivity of turn-taking also calls for language-dependent optimization. Given that NNS experience faster and smoother processing in their native language (L1), comprehensive and explanatory content, such as clarifications and summaries, could be displayed in L1 to further reduce the cognitive load during turn-taking. In contrast, suggestive content such as ideas and concepts expected to be used could be displayed in the second language for instant implementation.

6 Conclusion and Limitations

This study evaluates scaffolding turn-taking with proactive HCI systems in turn-taking. We conducted a WoZ study with 27 NNS college students using XPLAIN, a proactive system embedded in virtual meetings to generate clarifications, suggestions, and summaries to scaffold both speech comprehension and production in real-time. Participants validated the effectiveness of XPLAIN in supporting different communication gaps during turn-taking. Our findings call for proactive systems adaptive to individual differences in communication needs, personal background, and prior experience with AI to reduce language-related anxiety and enhance collaborative engagement in virtual environments. Incorporating these sociocultural considerations into user-centered design, we stress the balance of information delivery with optimal processing and create an experience with autonomy, inclusivity, and social comfort for diverse user populations. Future studies should test different age groups or occupations to generalize the findings beyond educational settings.

Acknowledgments

The authors thank anonymous reviewers for their valuable feedback. We also appreciate the great insights and support from the mentors and advisors of the Cornell Entrepreneurship Lab program: Nate Cook, Donnie Hampton, Clyde McKee, Kenneth S. Rother, Gregory Roy, Brad Treat, Kayla Wooley, and Jacob Wright. We also thank

our wonderful research assistants, Jay Mu-Chieh Huang, Teresa Lian, and Sabrina Lin, for their effort in designing and running this study.

References

- [1] Hussein A. Abbass. 2019. Social Integration of Artificial Intelligence: Functions, Automation Allocation Logic and Human-Autonomy Trust. *Cognitive Computation* 11, 2 (2019), 159–171. doi:10.1007/s12559-018-9619-0
- [2] Rhona M. Amos and Martin J. Pickering. 2020. A theory of prediction in simultaneous interpreting. *Bilingualism: Language and Cognition* 23, 4 (2020), 706–715. doi:10.1017/S1366728919000671
- [3] Salvatore Andolina, Valeria Orso, Hendrik Schneider, Khalil Klouche, Tuukka Ruotsalo, Luciano Gamberini, and Giulio Jacucci. 2018. Investigating Proactive Search Support in Conversations. In *Proceedings of the 2018 Designing Interactive Systems Conference*, ACM, Hong Kong China, 1295–1307. doi:10.1145/3196709.3196734
- [4] Kenneth C. Arnold, Krysta Chauncey, and Krzysztof Z. Gajos. 2020. Predictive text encourages predictable writing. In *Proceedings of the 25th International Conference on Intelligent User Interfaces*, ACM, Cagliari Italy, 128–138. doi:10.1145/3377325.3377523
- [5] Kenneth C. Arnold, Krzysztof Z. Gajos, and Adam T. Kalai. 2016. On Suggesting Phrases vs. Predicting Words for Mobile Text Composition. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology*, ACM, Tokyo Japan, 603–608. doi:10.1145/2984511.2984584
- [6] Advait Bhat, Saaket Agashe, and Anirudha Joshi. 2021. How do people interact with biased text prediction models while writing?. In *Proceedings of the first workshop on bridging human-computer interaction and natural language processing*, 116–121.
- [7] Heather Bortfeld, Silvia D. Leon, Jonathan E. Bloom, Michael F. Schober, and Susan E. Brennan. 2001. Disfluency Rates in Conversation: Effects of Age, Relationship, Topic, Role, and Gender. *Language and Speech* 44, 2 (2001), 123–147. doi:10.1177/00238309010440020101
- [8] Patricia A. Broen and Gerald M. Siegel. 1972. Variations in Normal Speech Disfluencies. *Language and Speech* 15, 3 (1972), 219–231. doi:10.1177/002383097201500302
- [9] Zana Bućinca, Maja Barbara Malaya, and Krzysztof Z. Gajos. 2021. To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI in AI-assisted Decision-making. In *Proceedings of the ACM on Human-Computer Interaction*, Vol. 5, 1–21. doi:10.1145/3449287
- [10] Sara Bögels and Francisco Torreira. 2015. Listeners use intonational phrase boundaries to project turn ends in spoken interaction. *Journal of Phonetics* 52 (2015), 46–57. doi:10.1016/j.jwocn.2015.04.004
- [11] Wanling Cai, Yucheng Jin, and Li Chen. 2022. Impacts of Personal Characteristics on User Trust in Conversational Recommender Systems. In *CHI Conference on Human Factors in Computing Systems*, ACM, New Orleans LA USA, 1–14. doi:10.1145/3491102.3517471
- [12] Jon W. Carr, Kenny Smith, Hannah Cornish, and Simon Kirby. 2017. The Cultural Evolution of Structured Languages in an Open-Ended, Continuous World. *Cognitive Science* 41, 4 (2017), 892–923. doi:10.1111/cogs.12371
- [13] Herbert H. Clark and Susan E. Brennan. 1991. *Grounding in communication*. American Psychological Association, Washington, 127–149. <http://content.apa.org/books/10096-006>
- [14] Ruth E. Corps, Abigail Crossley, Chiara Gambi, and Martin J. Pickering. 2018. Early preparation during turn-taking: Listeners use content predictions to determine what to say but not when to say it. *Cognition* 175 (2018), 77–95. doi:10.1016/j.cognition.2018.01.015
- [15] Ruth E. Corps, Meijian Liao, and Martin J. Pickering. 2023. Evidence for two stages of prediction in non-native speakers: A visual-world eye-tracking study. *Bilingualism: Language and Cognition* 26, 1 (2023), 231–243. doi:10.1017/S1366728922000499
- [16] Mark Davies. 2008. The Corpus of Contemporary American English. www.english-corpora.org/coca/
- [17] Jan P. de Ruiter, Adrian Bangerter, and Paula Dings. 2012. The Interplay Between Gesture and Speech in the Production of Referring Expressions: Investigating the Tradeoff Hypothesis. *Topics in Cognitive Science* 4, 2 (2012), 232–248. doi:10.1111/j.1756-8765.2012.01183.x
- [18] Yang Deng, Wenqiang Lei, Minlie Huang, and Tat-Seng Chua. 2023. Rethinking Conversational Agents in the Era of LLMs: Proactivity, Non-collaborativity, and Beyond. In *Proceedings of the Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region*, ACM, Beijing China, 298–301. doi:10.1145/3624918.3629548
- [19] Yang Deng, Wenqiang Lei, Wai Lam, and Tat-Seng Chua. 2023. A Survey on Proactive Dialogue Systems: Problems, Methods, and Prospects. arXiv:2305.02750 [cs.CL] <https://arxiv.org/abs/2305.02750>
- [20] Yang Deng, Lizi Liao, Zhonghua Zheng, Grace Hui Yang, and Tat-Seng Chua. 2024. Towards Human-centered Proactive Conversational Agents. In *Proceedings of the*

- 47th International ACM SIGIR Conference on Research and Development in Information Retrieval (Washington DC, USA) (SIGIR '24). Association for Computing Machinery, New York, NY, USA, 807–818. doi:10.1145/3626772.3657843
- [21] Finale Doshi-Velez and Been Kim. 2017. Towards A Rigorous Science of Interpretable Machine Learning. arXiv:1702.08608 [stat.ML] <https://arxiv.org/abs/1702.08608>
- [22] Wen Duan, Naomi Yamashita, and Susan R. Fussell. 2019. Increasing Native Speakers' Awareness of the Need to Slow Down in Multilingual Conversations Using a Real-Time Speech Speedometer. In *Proceedings of the ACM on Human-Computer Interaction*, Vol. 3. 1–25. doi:10.1145/3359273
- [23] Wen Duan, Naomi Yamashita, Yoshinari Shirai, and Susan R. Fussell. 2021. Bridging Fluency Disparity between Native and Nonnative Speakers in Multilingual Multiparty Collaboration Using a Clarification Agent. In *Proceedings of the ACM on Human-Computer Interaction*, Vol. 5. 1–31. doi:10.1145/3479579
- [24] Fernanda Ferreira and Karl G.D. Bailey. 2004. Disfluencies and human language comprehension. *Trends in Cognitive Sciences* 8, 5 (2004), 231–237. doi:10.1016/j.tics.2004.03.011
- [25] Ge Gao, Bin Xu, David C. Hau, Zheng Yao, Dan Cosley, and Susan R. Fussell. 2015. Two is Better Than One: Improving Multilingual Collaboration by Giving Two Machine Translation Outputs. In *Proceedings of the 18th ACM Conference on Computer Supported Cooperative Work & Social Computing*. ACM, Vancouver BC Canada, 852–863. doi:10.1145/2675133.2675197
- [26] Barney G. Glaser and Anselm L. Strauss. 2017. *The discovery of grounded theory: strategies for qualitative research*. Routledge, London New York.
- [27] Lauren V. Hadley and John F. Culling. 2022. Timing of head turns to upcoming talkers in triadic conversation: Evidence for prediction of turn ends and interruptions. *Frontiers in Psychology* 13 (2022), 1061582. doi:10.3389/fpsyg.2022.1061582
- [28] Wanqing He and Susan R. Fussell. 2025. Difference in the Cognitive Mechanism of Predictive Processing in Computer-Mediated Communication: A Comparison Study of L2 Speakers. In *Proceedings of the 43rd Annual Conference of the Cognitive Science Society*. San Francisco California USA.
- [29] Wanqing He, Calen C. MacDonald, Yejoon Yoo, Marcos Eizayaga, Ryun Shim, Lev D. Katreczko, and Susan R. Fussell. 2025. Exploring Content Predictability in Turn-Taking Through Different Computer-Mediated Communications. In *Proceedings of the 31st International Conference on Computational Linguistics*, Owen Rambow, Leo Wanner, Marianna Apidianaki, Hend Al-Khalifa, Barbara Di Eugenio, and Steven Schockaert (Eds.). Association for Computational Linguistics, Abu Dhabi, UAE, 7949–7962. <https://aclanthology.org/2025.coling-main.532/>
- [30] Jess Hohenstein and Malte Jung. 2018. AI-Supported Messaging: An Investigation of Human-Human Text Conversation with AI Support. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems*. ACM, Montreal QC Canada, 1–6. doi:10.1145/3170427.3188487
- [31] Falk Huettig. 2015. Four central questions about prediction in language processing. *Brain Research* 1626 (2015), 118–135. doi:10.1016/j.brainres.2015.02.014
- [32] Maurice Jakesch, Advait Bhat, Daniel Buschek, Lior Zalmanson, and Mor Naaman. 2023. Co-Writing with Opinionated Language Models Affects Users' Views. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM, Hamburg Germany, 1–15. doi:10.1145/3544548.3581196
- [33] Matthias Kraus, Nicolas Wagner, and Wolfgang Minker. 2020. Effects of Proactive Dialogue Strategies on Human-Computer Trust. In *Proceedings of the 28th ACM Conference on User Modeling, Adaptation and Personalization*. ACM, Genoa Italy, 107–116. doi:10.1145/3340631.3394840
- [34] Emily Kuang, Minghao Li, Mingming Fan, and Kristen Shinohara. 2024. Enhancing UX Evaluation Through Collaboration with Conversational AI Assistants: Effects of Proactive Dialogue and Timing. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 3, 16 pages. doi:10.1145/3613904.3642168
- [35] Gina R. Kuperberg and T. Florian Jaeger. 2016. What do we mean by prediction in language comprehension? *Language, cognition and neuroscience* 31, 1 (2016), 32–59. doi:10.1080/23273798.2015.1102299
- [36] Jean Lave and Etienne Wenger. 1991. *Situated learning: legitimate peripheral participation*. Cambridge University Press, Cambridge [England] ; New York.
- [37] Shiri LevAri, Emily Ho, and Boaz Keysar. 2018. The Unforeseen Consequences of Interacting With NonNative Speakers. *Topics in Cognitive Science* 10, 4 (2018), 835–849. doi:10.1111/tops.12325
- [38] Stephen C. Levinson. 2016. Turn-taking in Human Communication – Origins and Implications for Language Processing. *Trends in Cognitive Sciences* 20, 1 (2016), 6–14. doi:10.1016/j.tics.2015.10.010
- [39] Xiaoyan Li, Naomi Yamashita, Wen Duan, Yoshinari Shirai, and Susan R. Fussell. 2023. Improving Non-Native Speakers' Participation with an Automatic Agent in Multilingual Groups. *Proceedings of the ACM on Human-Computer Interaction* 7 (2023), 1–28. doi:10.1145/3567562
- [40] Lizi Liao, Grace Hui Yang, and Chirag Shah. 2023. Proactive Conversational Agents in the Post-ChatGPT World. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*. ACM, Taipei Taiwan, 3452–3455. doi:10.1145/3539618.3594250
- [41] Hajin Lim, Dan Cosley, and Susan R. Fussell. 2022. Understanding Cross-lingual Pragmatic Misunderstandings in Email Communication. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW1 (2022), 1–32. doi:10.1145/3512976
- [42] Yuhuan Liu, Aadit Shah, Jordan Ackerman, and Manaswi Saha. 2025. Exploring the Design Space of Real-time LLM Knowledge Support Systems: A Case Study of Jargon Explanations. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–20. doi:10.1145/3706598.3714262
- [43] Ewa Luger and Abigail Sellen. 2016. "Like Having a Really Bad PA": The Gulf between User Expectation and Experience of Conversational Agents. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. ACM, San Jose California USA, 5286–5297. doi:10.1145/2858036.2858288
- [44] Lilla Magyari. 2022. *Predictions in Conversation*. Vol. 11. Springer International Publishing, Cham, 59–75. https://link.springer.com/10.1007/978-3-030-66175-5_5
- [45] Lilla Magyari and J. P. de Ruiter. 2012. Prediction of Turn-Ends Based on Anticipation of Upcoming Words. *Frontiers in Psychology* 3 (2012), 376. doi:10.3389/fpsyg.2012.00376
- [46] Lilla Magyari, Jan P. De Ruiter, and Stephen C. Levinson. 2017. Temporal Preparation for Speaking in Question-Answer Sequences. *Frontiers in Psychology* 8 (2017). <https://www.frontiersin.org/articles/10.3389/fpsyg.2017.00211>
- [47] Mei-Hua Pan, Naomi Yamashita, and Hao-Chuan Wang. 2017. Task Rebalancing: Improving Multilingual Communication with Native Speakers-Generated Highlights on Automated Transcripts. In *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing*. ACM, Portland Oregon USA, 310–321. doi:10.1145/2998181.2998304
- [48] Zhenhui Peng, Yunhwan Kwon, Jiaan Lu, Ziming Wu, and Xiaojuan Ma. 2019. Design and Evaluation of Service Robot's Proactivity in Decision-Making Support Process. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, Glasgow Scotland UK, 1–13. doi:10.1145/3290605.3300328
- [49] Martin J. Pickering and Simon Garrod. 2007. Do people use language production to make predictions during comprehension? *Trends in Cognitive Sciences* 11, 3 (2007), 105–110. doi:10.1016/j.tics.2006.12.002
- [50] Martin J. Pickering and Simon Garrod. 2013. An integrated theory of language production and comprehension. *Behavioral and Brain Sciences* 36, 4 (2013), 329–347. doi:10.1017/S0140525X12001495
- [51] Peinuan Qin, Zicheng Zhu, Naomi Yamashita, Yitian Yang, Keita Suga, and Yi-Chieh Lee. 2025. AI-Based Speaking Assistant: Supporting Non-Native Speakers' Speaking in Real-Time Multilingual Communication. arXiv:2505.01678 [cs.HC] <https://arxiv.org/abs/2505.01678>
- [52] Evan F. Risko and Sam J. Gilbert. 2016. Cognitive Offloading. *Trends in Cognitive Sciences* 20, 9 (2016), 676–688. doi:10.1016/j.tics.2016.07.002
- [53] Rachel Ryskin and Mante S. Nieuwland. 2023. Prediction during language comprehension: what is next? *Trends in Cognitive Sciences* 27, 11 (2023), 1032–1052. doi:10.1016/j.tics.2023.08.003
- [54] Michael Tomasello. 2008. *Origins of human communication*. Number 2008 in The Jean Nicod lectures. MIT Press, Cambridge, Mass.
- [55] Francisco Torreira, Sara Bögel, and Stephen C. Levinson. 2015. Breathing for answering: the time course of response planning in conversation. *Frontiers in Psychology* 6 (2015). doi:10.3389/fpsyg.2015.00284
- [56] James P. Trujillo, Stephen C. Levinson, and Judith Holler. 2021. Visual Information in Computer-Mediated Interaction Matters: Investigating the Association Between the Availability of Gesture and Turn Transition Timing in Conversation. In *Human-Computer Interaction. Design and User Experience Case Studies (Lecture Notes in Computer Science)*, Masaaki Kurosu (Ed.). Springer International Publishing, Cham, 643–657. doi:10.1007/978-3-030-78468-3_44
- [57] Joseph B. Walther, Brandon Van Der Heide, Artemio Ramirez, Judee K. Burgoon, and Jorge Peña. 2015. *Interpersonal and Hyperpersonal Dimensions of Computer-Mediated Communication* (1 ed.). Wiley, 1–22. <https://dl.acm.org/10.1002/9781118426456.ch1>
- [58] Ben Wang, Jiqun Liu, Jamshed Karimnazarov, and Nicolas Thompson. 2024. Task Supportive and Personalized Human-Large Language Model Interaction: A User Study. In *Proceedings of the 2024 ACM SIGIR Conference on Human Information Interaction and Retrieval*. ACM, Sheffield United Kingdom, 370–375. doi:10.1145/3627508.3638344
- [59] Tessa Warren and Michael Walsh Dickey. 2021. The use of linguistic and world knowledge in language processing. *Language and Linguistics Compass* 15, 4 (2021), e12411. doi:10.1111/lnc3.12411
- [60] Naomi Yamashita, Andy Echenique, Toru Ishida, and Ari Hautasaari. 2013. Lost in transmittance: how transmission lag enhances and deteriorates multilingual collaboration. In *Proceedings of the 2013 Conference on Computer Supported Cooperative Work* (San Antonio, Texas, USA) (CSCW '13). Association for Computing Machinery, New York, NY, USA, 923–934. doi:10.1145/2441776.2441881
- [61] Rui Yan and Dongyan Zhao. 2018. Smarter Response with Proactive Suggestion: A New Generative Neural Conversation Paradigm. In *Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence*. International Joint Conferences on Artificial Intelligence Organization, Stockholm, Sweden, 4525–4531. doi:10.24963/ijcai.2018/629